# Exercise Set 9: Integration of Differential Forms

*Differential forms were built to be integrated. A $k$-form is precisely the kind of object that can be integrated over a $k$-dimensional oriented manifold in a coordinate-independent way. The master theorem is Stokes' theorem: $\int_M d\omega = \int_{\partial M} \omega$. It unifies the fundamental theorem of calculus, Green's theorem, and the divergence theorem into a single geometric statement. Its deepest consequence is topological: a closed form with nonzero integral cannot be exact and this is the first glimpse of de Rham cohomology.*

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## Recall of Definitions and Examples

### Orientation

**R1.** Write the definition of an **oriented manifold**. What does it mean for an atlas to be **oriented**? What is the condition on the Jacobians of the transition maps?

**R2.** For each of the following manifolds, state whether it is orientable. If yes, describe a choice of orientation.

(a) $\mathbb{R}^n$

(b) $S^n$ for any $n \geq 1$

(c) $T^2 = S^1 \times S^1$

(d) The Möbius strip

(e) $\mathbb{R}P^2$

**R3.** What is the **boundary** $\partial M$ of a manifold with boundary? Give the boundary of:

(a) $[0, 1]$

(b) $[0, 1]^2$

(c) The closed disk $D^2 = \{(x, y) : x^2 + y^2 \leq 1\}$

(d) $S^2$ (as a manifold without boundary)

(e) The upper hemisphere $D^2_+ = \{(x, y, z) \in S^2 : z \geq 0\}$

### Integration of forms

**R4.** Write the definition of $\int_M \omega$ for an $n$-form $\omega$ on an oriented $n$-manifold $M$. How does a chart $(U, \phi)$ reduce this to an ordinary integral on $\mathbb{R}^n$? Why does the orientation ensure the result is well-defined (independent of charts)?

**R5.** State the **change of variables formula**: if $f : M \to N$ is an orientation-preserving diffeomorphism and $\omega \in \Omega^n(N)$, then

$$\int_N \omega = \int_M f^*\omega.$$

How does this generalise the change of variables formula from multivariable calculus?

**R6. Stokes' theorem.** State the theorem precisely: for an oriented compact $n$-manifold $M$ with boundary $\partial M$ (with the induced orientation), and $\omega \in \Omega^{n-1}(M)$:

$$\int_M d\omega = \int_{\partial M} \omega.$$

What happens when $M$ has no boundary?

**R7.** State the following as special cases of Stokes' theorem:

(a) **Fundamental theorem of calculus**: $\int_a^b f'(x)\,dx = f(b) - f(a)$

(b) **Green's theorem** on a region $D \subset \mathbb{R}^2$

(c) **Divergence theorem** on a region $V \subset \mathbb{R}^3$

(d) **Classical Stokes' theorem** for a surface $S \subset \mathbb{R}^3$

For each, identify: what is $M$? What is $\partial M$? What is $\omega$?

**R8.** State the two key consequences of Stokes' theorem:

(a) If $\omega = d\alpha$ is exact and $M$ is closed (no boundary), then $\int_M \omega = ?$

(b) If $\omega$ is closed ($d\omega = 0$) and $\int_M \omega \neq 0$, then $\omega$ is ? (exact or not exact?)

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# Theory

**T1. (Integration is chart-independent.)** Let $\omega$ be an $n$-form on an oriented $n$-manifold $M$, and let $(U, \phi)$ and $(V, \psi)$ be two overlapping oriented charts. Show that the two local integrals agree on $U \cap V$:

$$\int_{\phi(U\cap V)} (\phi^{-1})^*\omega = \int_{\psi(U\cap V)} (\psi^{-1})^*\omega.$$

*Hint: use the change of variables formula and the fact that the transition map $\psi \circ \phi^{-1}$ has positive Jacobian determinant (orientation-preserving).*

**T2. (Stokes implies exact forms integrate to zero.)** Let $M$ be a compact oriented manifold *without boundary* (a **closed manifold**). Show that for any $\omega = d\alpha$:

$$\int_M \omega = 0.$$

Use this to prove: if $\int_M \omega \neq 0$, then $\omega$ is not exact.

**T3. (Closed forms and the topology of the domain.)** Let $\omega$ be a closed $n$-form on a manifold $M$.

(a) If $M$ is contractible (e.g. $M = \mathbb{R}^n$), state what happens to $\omega$ (Poincaré lemma: every closed form on $\mathbb{R}^n$ is exact).

(b) If $M$ is not contractible (e.g. $M = S^n$ or $M = \mathbb{R}^n \setminus \{0\}$), give an example where $\omega$ is closed but not exact.

(c) This shows: *the failure of closed forms to be exact is a topological property of $M$.* The set of closed forms that are not exact is what de Rham cohomology measures. We will study this in Set 10.

**T4. (Induced orientation on the boundary.)** The boundary $\partial M$ inherits an orientation from $M$: for an $n$-manifold $M$ with boundary, if $(e_1, \ldots, e_n)$ is a positively oriented frame on $M$ near $\partial M$, and $\nu$ is the outward normal, then $(e_1, \ldots, e_{n-1})$ is the induced orientation on $\partial M$ such that $(\nu, e_1, \ldots, e_{n-1})$ is positively oriented on $M$.

Apply this to find the induced orientations:

(a) On $\partial[0,1] = \{0\} \cup \{1\}$: which point gets $+$ and which $-$?

(b) On $\partial D^2 = S^1$: which direction (clockwise or counterclockwise) is the induced orientation?

(c) On $\partial([0,1]^2)$: describe the induced orientation on each of the four edges.

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# Computation

## Basic integrals

**C1. (Integration on $[0,1]$ and $S^1$.)**

(a) Compute $\int_{[0,1]} f'(x)\,dx$ for $f(x) = x^2$. Verify using Stokes: $\int_{[0,1]} df = f(1) - f(0)$.

(b) Compute $\int_{S^1} d\theta$ where $\theta$ is the angle coordinate. Is $d\theta$ exact on $S^1$?

(c) Let $\omega = \sin\theta\, d\theta$ on $S^1$. Compute $\int_{S^1} \omega$. Is $\omega$ exact?
*Hint: find $f$ with $df = \omega$.*

(d) Let $\gamma : [0, 2\pi] \to S^1$, $\gamma(t) = (\cos t, \sin t)$. Compute $\int_{S^1} x\,dy$ using the parametrisation $\gamma$.

**C2. (Integration on $[0,1]^2$ and $D^2$.)**

(a) Compute $\int_{[0,1]^2} dx \wedge dy$.

(b) Compute $\int_{[0,1]^2} xy\, dx \wedge dy$.

(c) Let $\omega = x\, dy - y\, dx$. Compute $\int_{D^2} d\omega$ and $\int_{\partial D^2} \omega$ separately, then verify Stokes' theorem.

(d) Compute $\int_{D^2} (x^2 + y^2)\, dx \wedge dy$ using polar coordinates.

**C3. (Integration on $S^2$.)**

(a) Using spherical coordinates $(\theta, \varphi)$, compute the area of $S^2$:

$$\int_{S^2} \sin\theta\, d\theta \wedge d\varphi.$$

(b) Compute $\int_{S^2} z \sin\theta\, d\theta \wedge d\varphi$ where $z = \cos\theta$.

(c) Let $\omega = x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy$ be the area form on $S^2 \subset \mathbb{R}^3$. Verify that $\int_{S^2} \omega = 4\pi$ using the parametrisation.

(d) Using the divergence theorem, compute $\int_{S^2} \omega$ by instead computing $\int_{B^3} d\omega$ where $B^3$ is the unit ball. Verify both methods give the same answer.

**C4. (Verifying Stokes on explicit examples.)**

(a) Let $M = [0,1] \times [0,1]$ and $\omega = xy\, dx + x^2\, dy$. Compute $\int_M d\omega$ directly, then compute $\int_{\partial M} \omega$ by integrating along the four edges. Verify they agree.

(b) Let $S$ be the upper hemisphere $\{z \geq 0\} \cap S^2$ with boundary $\partial S = S^1$ (the equator) oriented counterclockwise. Let $\omega = -y\, dx + x\, dy$. Compute $\int_S d\omega$ and $\int_{S^1} \omega$ separately. Verify Stokes' theorem.

(c) Let $V$ be the unit ball $B^3 \subset \mathbb{R}^3$ and $\omega = x^2\, dy \wedge dz + y^2\, dz \wedge dx + z^2\, dx \wedge dy$. Compute $\int_{S^2} \omega$ (the boundary integral) using Stokes, by first computing $d\omega$.

**C5. (The angle form and its integral.)**

The angle form on $\mathbb{R}^2 \setminus \{0\}$ is:

$$\omega = \frac{-y\, dx + x\, dy}{x^2 + y^2}.$$

(a) Show $d\omega = 0$ (closed).

(b) Compute $\oint_{S^1} \omega$ where $S^1$ is the unit circle parametrised by $(\cos t, \sin t)$, $t \in [0, 2\pi]$.

(c) Show $\omega$ is not exact on $\mathbb{R}^2 \setminus \{0\}$ using the result of (b) and Stokes' theorem.

(d) On $\mathbb{R}^2$ (no puncture), is $\omega$ exact? If so, find $f$ with $df = \omega$.

(e) Compute $\oint_\gamma \omega$ where $\gamma$ is a circle of radius 2 centred at $(3, 0)$ (not enclosing the origin). Explain the result using topology.

**C6. (Integration and pullback.)**

(a) Let $f : S^1 \to S^1$ be $f(\theta) = 2\theta$ (the double cover). Compute $\int_{S^1} f^*(d\theta)$.

(b) Let $\iota : S^1 \hookrightarrow \mathbb{R}^2$ be the inclusion. Compute $\int_{S^1} \iota^*(x\,dy)$.

(c) Let $g : [0,1] \to S^1$, $g(t) = (\cos 2\pi t, \sin 2\pi t)$. Show $g^*(d\theta) = 2\pi\,dt$ and compute $\int_{[0,1]} g^*(d\theta)$.

(d) Verify: $\int_{S^1} d\theta = \int_{[0,1]} g^*(d\theta)$. This is the change of variables formula in action.

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# Physics: Stokes' Theorem, Gauss's Law, and Topology

*Stokes' theorem $\int_M d\omega = \int_{\partial M} \omega$ is the geometric foundation of every conservation law and every flux integral in physics. Gauss's law the most fundamental law of electrostatics is exactly Stokes' theorem applied to a closed but non-exact 2-form. The fact that the integral is nonzero is not a property of the field: it is a property of the topology of the domain $\mathbb{R}^3 \setminus \{0\}$. In this section we make this precise, following Nawata (HW3, Problem 3).*

## The generator of $H^n_{dR}(S^n)$

**P1. (The canonical volume form on $S^n$.)**

Define the $n$-form on $\mathbb{R}^{n+1} \setminus \{0\}$:

$$\omega = \frac{1}{|x|^{n+1}} \sum_{i=0}^{n} (-1)^i x^i\, dx^0 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n,$$

where $\widehat{dx^i}$ means $dx^i$ is omitted, and $|x|^2 = \sum_i (x^i)^2$.

(a) For $n = 1$: write $\omega$ explicitly on $\mathbb{R}^2 \setminus \{0\}$ with coordinates $(x^0, x^1) = (x, y)$. Compare with the angle form from C5. Are they the same?

(b) For $n = 2$: write $\omega$ explicitly on $\mathbb{R}^3 \setminus \{0\}$ with coordinates $(x^0, x^1, x^2) = (x, y, z)$.

(c) Show that $d\omega = 0$ on $\mathbb{R}^{n+1} \setminus \{0\}$.
*Hint: compute $d\omega$ by the Leibniz rule. The numerator contributes $\sum_i (-1)^i dx^i \wedge dx^0 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n = (n+1) dx^0 \wedge \cdots \wedge dx^n$. The denominator contributes terms that cancel with these by the quotient rule.*

(d) Let $\iota : S^n \hookrightarrow \mathbb{R}^{n+1}$ be the inclusion of the unit sphere. Show that $\iota^*\omega$ is the standard volume form on $S^n$.

(e) Compute $\int_{S^2} \iota^*\omega$ for $n = 2$.
*Hint: use spherical coordinates $x = \sin\theta\cos\varphi$, $y = \sin\theta\sin\varphi$, $z = \cos\theta$ and the fact that $|x| = 1$ on $S^2$.*

(f) Show that $\iota^*\omega$ is **not exact** on $S^n$ by using the result of (e) and Stokes' theorem. Conclude: $\iota^*\omega$ represents a nonzero cohomology class in $H^n_{dR}(S^n)$. In fact, $H^n_{dR}(S^n) \cong \mathbb{R}$, generated by $[\iota^*\omega]$.

**P2. (Gauss's law as Stokes' theorem.)**

A point charge $Q$ at the origin of $\mathbb{R}^3$ creates the electric field:

$$\mathbf{E}(x, y, z) = \frac{Q}{4\pi} \cdot \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}.$$

Define the associated 2-form (the **electric flux form**):

$$\Phi_E = \frac{Q}{4\pi} \cdot \frac{x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy}{(x^2 + y^2 + z^2)^{3/2}}.$$

(a) Show that $\Phi_E = \frac{Q}{4\pi}\iota^*\omega$ where $\omega$ is the $n = 2$ form from P1. What is the constant of proportionality?

(b) Show $d\Phi_E = 0$ on $\mathbb{R}^3 \setminus \{0\}$. This is the statement: **away from the charge, there are no sources**. In vector calculus notation: $\nabla \cdot \mathbf{E} = 0$ for $x \neq 0$.

(c) Compute $\int_{S^2_r} \Phi_E$ where $S^2_r$ is a sphere of radius $r$ centred at the origin. Show the result is independent of $r$. This is **Gauss's law**.

(d) Now let $\Sigma$ be any closed surface surrounding the origin (not necessarily a sphere). Using Stokes' theorem, show:

$$\int_\Sigma \Phi_E = Q.$$

*Hint: since $d\Phi_E = 0$ away from the origin, use Stokes on the region between $\Sigma$ and a small sphere $S^2_\epsilon$.*

(e) What happens if $\Sigma$ does *not* surround the origin? Compute $\int_\Sigma \Phi_E$ in this case.

(f) **Key observation.** The charge $Q$ is detected by $\int_\Sigma \Phi_E$. This integral is:

- Independent of the shape of $\Sigma$ (only depends on whether it surrounds the charge)
- Independent of the metric (it's a form integral, not a vector calculus operation)
- Nonzero because $\Phi_E$ is closed but not exact on $\mathbb{R}^3 \setminus \{0\}$

The charge is a **topological invariant** of the field configuration it measures the cohomology class $[\Phi_E] \in H^2_{dR}(\mathbb{R}^3 \setminus \{0\})$. Write one sentence connecting this to the winding number from Set 4.

**P3. (Stokes' theorem unifies all of classical physics.)**

Each of the following is Stokes' theorem $\int_M d\omega = \int_{\partial M} \omega$ in disguise. For each, identify $M$, $\partial M$, and $\omega$.

(a) **Faraday's law** (from Maxwell's equations in Set 8):

$$\oint_{\partial S} \mathbf{E} \cdot d\ell = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{A}.$$

(b) **Ampère's law**:

$$\oint_{\partial S} \mathbf{B} \cdot d\ell = \int_S \mathbf{J} \cdot d\mathbf{A} + \frac{\partial}{\partial t} \int_S \mathbf{E} \cdot d\mathbf{A}.$$

(c) **Conservation of charge**: in forms language, $dJ = 0$ (from Set 8). Show this is equivalent to $\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$, and that integrating over a volume $V$ gives:

$$\frac{d}{dt} \int_V \rho \, d^3x = -\oint_{\partial V} \mathbf{J} \cdot d\mathbf{A}.$$

This says: the rate of change of charge in $V$ equals minus the flux out through $\partial V$. This is Stokes applied to $J$.

**P4. (Preview of de Rham cohomology.)**

We have now seen several closed forms that are not exact:

- $d\theta$ on $S^1$ (Set 4, Set 8): $\oint_{S^1} d\theta = 2\pi \neq 0$
- $\iota^*\omega$ on $S^2$ (P1 above): $\int_{S^2} \iota^*\omega = 4\pi \neq 0$
- The electric flux form $\Phi_E$ on $\mathbb{R}^3 \setminus \{0\}$ (P2 above)

(a) In each case, what is the topological feature of the domain that prevents the form from being exact? (A hole? A puncture? A non-contractible cycle?)

(b) The integral $\int_M \omega$ for a closed form $\omega$ depends only on the **homology class** of the cycle $M$ not on the specific representative. Explain what this means for Gauss's law: why does the shape of the Gaussian surface not matter?

(c) In Set 10 we will define de Rham cohomology $H^k_{dR}(M) = \ker d / \operatorname{im} d$ precisely. Based on your work in this set, what do you expect $H^1_{dR}(S^1)$ and $H^2_{dR}(S^2)$ to be?

(d) **The running thread.** In Set 4 we used the winding number to detect holes in $\mathbb{R}^2 \setminus \{0\}$. In this set we detected the hole in $\mathbb{R}^3 \setminus \{0\}$ using $\int_{S^2} \Phi_E = Q \neq 0$. In Set 10 we will make this precise: de Rham cohomology is the algebraic machine that counts and classifies these holes. Write one sentence describing what $H^k_{dR}(M)$ should measure, based on everything you have seen so far.

Gauss's law is exactly this: $\Phi_E$ is closed ($d\Phi_E = 0$) on $\mathbb{R}^3 \setminus \{0\}$, and $\int_{S^2} \Phi_E = Q \neq 0$, so $\Phi_E$ is not exact. The charge $Q$ is a topological invariant. This is physics detecting the topology of space.

In Set 10, we formalise this into de Rham cohomology:
$H^k_{dR}(M) = \{\text{closed } k\text{-forms}\}/\{\text{exact } k\text{-forms}\}$. Every integral $\int_M \omega$ for a closed form $\omega$ defines a linear functional on $H^k_{dR}(M)$ the pairing between cohomology and homology that is the content of de Rham's theorem.***Reflection.*** *Stokes' theorem* $\int_M d\omega = \int_{\partial M} \omega$ *is the fundamental theorem of calculus for differential forms on manifolds. Its topological consequence is stark: a closed form with nonzero integral over a closed manifold cannot be exact. This means the integral detects a topological obstruction the form "knows" about a hole in the space.*
*Gauss's law is exactly this:* $\Phi_E$ *is closed (*$d\Phi_E = 0$*) on* $\mathbb{R}^3 \setminus \{0\}$*, and* $\int_{S^2} \Phi_E = Q \neq 0$*, so* $\Phi_E$ *is not exact. The charge* $Q$ *is a topological invariant. This is physics detecting the topology of space.*
*In Set 10, we formalise this into de Rham cohomology:*
$H^k_{dR}(M) = \{\textit{closed } k\textit{-forms}\}/\{\textit{exact } k\textit{-forms}\}$. *Every integral* $\int_M \omega$ *for a closed form* $\omega$ *defines a linear functional on* $H^k_{dR}(M)$ *the pairing between cohomology and homology that is the content of de Rham's theorem.*